

silicon-based layered semiconductor that can produce electricity from sunlight. Just like all other types of battery cells, every solar cell has one positive back contact and one negative front contact.

When sunlight falls on a solar cell, some photons of the light that get absorbed have enough energy to exceed the energy gap between the valence band and conduction band as well as excite the electrons to break semiconductor bonds. Due to electrostatic force of the field across the junction, electrons migrate to the n-type side while holes move to the p-type side. This results in a potential difference across the terminals of the cell. A photovoltaic module or solar panel consists of multiple such cells connected in series to provide the required voltage output for a load.

Metrics affecting output of a solar panel

For any solar project to be successful, certain factors like the rating of a solar panel need to be chosen wisely. In case more voltage is required, more panels can be connected in the power generation system. The total system voltage of an array of panels is a product of the total number of panels and their individual voltage rating.

The parameters are a way to measure the efficiency of conversion of sunlight to electricity in a panel. For instance, if a tolerance value (in per cent) is mentioned in the power rating, it is used to specify the upper and lower bounds of power generation for the rated capacity. A few important

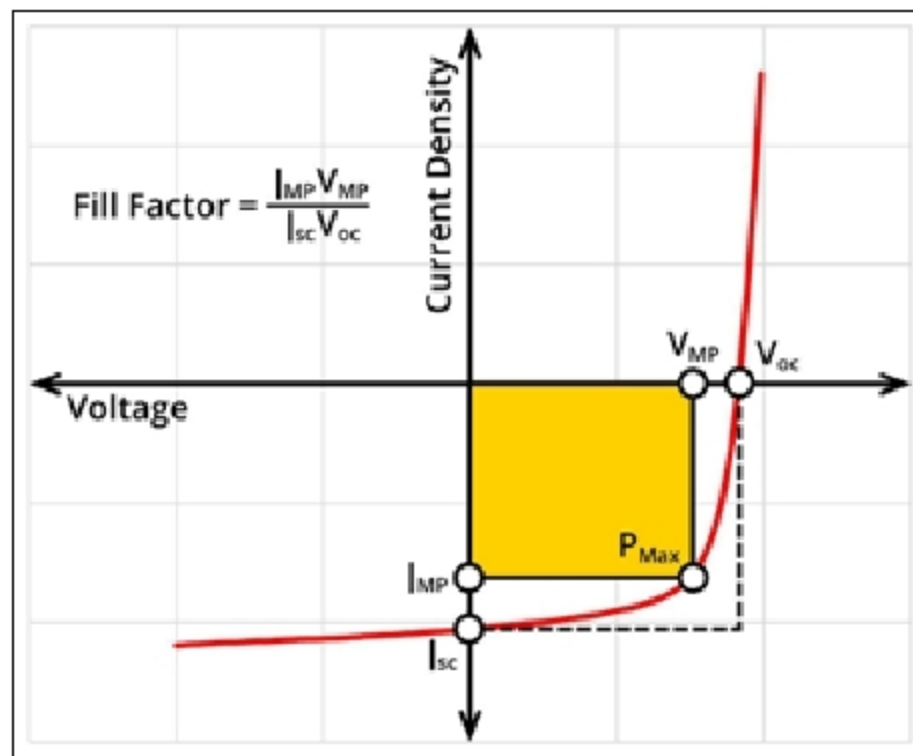


Fig. 2: Typical I-V curve of a solar cell (Credit: www.ossila.com)

parameters are discussed next.

Maximum power (P_{Max}). It is the maximum power (in watts) that can be obtained from a solar panel under standard conditions. The higher the value of maximum power, the greater is the load that it can handle. It can be calculated by multiplying current (I_{max}) with voltage value (V_{max}) at the maximum power point on the current-voltage (I-V) graph.

Voltage at maximum power (V_{MP}). The voltage of a solar panel (in volts) at peak efficiency when it is connected to the solar equipment (charge controller or inverter) under ambient conditions is known as the voltage at maximum power.

Current at maximum power (I_{MP}). The current generated by a solar panel (in amps) at peak efficiency when connected to the rest of the solar setup under ambient conditions is termed as the current at maximum power.

Open-circuit voltage (V_{oc}). As the

name suggests, V_{oc} is the voltage obtained across the terminals when no load is connected to the circuit or system. The current through the open circuit is always zero. V_{oc} can be simply measured using a voltmeter and is crucial in deciding the number of panels that can be connected in series.

Short-circuit current (I_{sc}). I_{sc} is the amount of current through solar panels when there is a short circuit between the positive and negative leads, that is, the voltage across the cells is zero. This is useful for determining the amount of current an inverter or con-

necter can withstand, thereby ensuring safety.

Efficiency (η). The percentage of absorbed solar energy that a solar panel can actually convert into electricity when in a complete setup under standard test conditions gives the efficiency of the panel. This is an essential measure in quantifying the performance of a panel; the higher the efficiency, the more is the amount of power generated. Factors like tilting the solar panels for better orientation to have full sunlight exposure and use of bipolar modules to produce power from both sides improve efficiency.

Fill factor (FF). Another metric to characterise the performance of a panel is the fill factor. The ratio of P_{max} and the value obtained on multiplying V_{oc} with I_{sc} gives the value of the fill factor.

Temperature coefficient. V_{oc} and I_{sc} decrease with increase in temperature (inversely proportional). The

TRACO POWER

www.tracopower.com

TMPW Series

Small Encapsulated AC/DC Power Supplies

- PCB or Chassis Mount styles
- 90-305 VAC Operation
- Class II Prepared (no Safety Ground)



- EN 55032 Class B Internal Filter
- IEC/EN/UL 62368-1 Approved

